

Does Real-Time AI Recommendation Reduce the Disposition Effect in Portfolio Management?

Experimental Economics | University of Pittsburgh | Graham Friedman

I. Research Question

Does receiving a real-time AI sell recommendation during an incentivized portfolio management task reduce the magnitude of the disposition effect, measured as the difference between the proportion of gains realized and the proportion of losses realized, in participants with finance training, compared to those who receive no AI recommendation?

II. Motivation and Related Literature

Portfolio managers sell their winners too early and hold their losers too long. Shefrin and Statman (1985) named this the disposition effect, and the mechanism is well established. Kahneman and Tversky's (1979) prospect theory shows that losses feel roughly twice as painful as equivalent gains feel good, so when a position is underwater, selling locks in pain that holding defers. Odean (1998) put a number on it: roughly 4.4% in annual foregone returns across more than 10,000 real brokerage accounts. Frazzini (2006) showed the same pattern in mutual fund managers. Professional training does not make it go away.

AI tools are now standard in portfolio management, marketed partly on the premise that algorithms do not feel loss aversion. That claim is credible for fully automated systems. It is untested for the more common case where a human manager receives an AI recommendation and still decides whether to act.

That premise deserves scrutiny before the experiment is worth running, because if an AI recommender were itself subject to the disposition effect it could not correct the human one, and the intervention would be pointless. The disposition effect is a reference-dependent bias: it requires a purchase price to anchor to and the felt pain of realizing a loss to defer. A recommender built on price momentum and pattern recognition, the kind described to subjects here, carries neither. It holds no position, realizes no loss, and has no reference point relative to which a paper loss becomes painful, so it can issue the sell signal that a loss-averse human suppresses. This is the mechanism we expect to operate, and it is a principled reason to believe an AI recommendation could succeed where professional training has failed. Training teaches managers what they should do but leaves the reference-dependent emotional response intact. An external recommendation supplies a judgment generated without that response in the first place. A system trained to imitate human traders could, of course, inherit the bias, which is precisely why the type of recommender matters and why we specify a momentum and pattern-recognition tool rather than a behavioral clone.

This distinction has direct consequences. If AI recommendations do shift disposition effect behavior, firms and regulators have a basis for building AI decision support into professional investment workflows, with implications for fiduciary standards and disclosure requirements. If they do not, resources being spent on these tools are not solving the behavioral problem they are marketed to solve. Either answer produces generalizable knowledge the industry currently lacks.

Guenther and Lordan (2023) showed that warning traders about the disposition effect before a trading task reduced their PGR minus PLR scores significantly across 193 traders, and Huang and Guenther (2024) found partial persistence of this effect over time. But pre-task

warnings change how someone enters a decision. A recommendation that arrives while you are already looking at a losing position operates differently: it has to compete with an active emotional response rather than shape the mindset before one forms. Logg et al. (2019) and Dietvorst et al. (2015) have studied how people respond to algorithmic versus human advice, but in forecasting contexts rather than incentivized portfolio management. No study has cleanly tested whether an in-the-moment AI recommendation shifts disposition effect behavior. That is the contribution of this experiment: not re-establishing a well-documented bias, but testing a specific, already-deployed intervention whose behavioral effect is currently unknown.

III. Experimental Design

We propose a framed field experiment (Harrison and List, 2004) implemented as a controlled portfolio-management simulation. Subjects manage a simulated portfolio of eight stocks over ten trading rounds, making buy, sell, or hold decisions each round after seeing updated prices. Stock names are fictional. Price sequences are pre-programmed in advance, drawn from historical equity volatility parameters, and held identical across both conditions, so that any difference in trading behavior reflects the treatment and not variation in the price environment. The experiment is implemented in oTree, an open-source platform standard in experimental economics for incentivized decision tasks of this kind.

At every round, subjects hold both gaining and losing positions simultaneously. This is a structural requirement: the disposition effect cannot be measured unless subjects face that simultaneous choice. In the AI condition, a notification banner appears each round: “AI Analysis: Recommend SELL, [Stock Name], [current P&L].” Subjects are told before starting: “Your portfolio interface includes an AI analysis feature based on price momentum and pattern recognition algorithms. Like all analytical tools, it is not always accurate, but it has shown strong

performance in historical backtesting.” In the no-AI condition, everything is identical but no banner appears.

Figure 2
Portfolio interface — AI condition (treatment)

• AI Analysis: Recommend SELL — Meridian Energy (+18.4%)

Stock	Purchase price	Current price	P&L	Action
Meridian Energy	\$42.00	\$49.72	+18.4%	Hold ▾
Vantage Capital	\$78.50	\$85.10	+8.4%	Hold ▾
Proxima Systems	\$61.20	\$53.85	-12.0%	Hold ▾
Crestline Holdings	\$33.40	\$31.10	-6.9%	Hold ▾
Apex Renewables	\$55.00	\$60.45	+9.9%	Hold ▾
NovaTech Inc.	\$90.00	\$83.20	-7.6%	Hold ▾
Summit Logistics	\$28.75	\$25.50	-11.3%	Hold ▾
Clearwater Global	\$44.10	\$47.30	+7.3%	Hold ▾

Round 4 of 10

Submit decisions

Highlighted row indicates the stock targeted by the AI recommendation this round. Subjects in the no-AI condition see the identical interface without the blue banner.

Recommendations are correct about 70% of the time, realistic for a financial AI tool. Making them always financially rational would let finance-trained subjects running through ten rounds notice the AI is always right, at which point we would be measuring instruction-following rather than behavioral change. A realistic error rate forces subjects to genuinely weigh each recommendation against their own judgment. Incorrect recommendations are distributed across both winning and losing positions across rounds, and some incorrect ones involve losing positions specifically, because we need rounds where the AI says sell a loser and the price subsequently recovers, so that the loss-side override pattern remains detectable even when the tool is occasionally wrong.

Under Harrison and List's (2004) classification this is a framed field experiment. The subject pool is finance students and junior professionals; the information they bring is working knowledge of portfolio management; the commodity is simulated equity positions with realistic price dynamics; the task is active portfolio management with buy, sell, and hold decisions each round; the stakes are tournament-based financial incentives; and the environment is a controlled lab setting. A conventional lab design with random undergraduates would not work here, because the disposition effect requires subjects who understand what a losing position means and feel something about holding it. Without that, the emotional mechanism does not activate.

A controlled simulation is the right instrument for this question. A natural field experiment inside an asset management firm would carry stronger external validity but could not isolate the effect: portfolio managers bring client mandates, firm investment policies, career incentives, and regulatory constraints into every decision, none of which can be controlled for, and randomization within a firm is impractical because managers share information and operate under shared mandates, so SUTVA violations would be immediate. Live trading of real stocks introduces a different confound, since real prices move on news and macroeconomic conditions, so a treated subject trading on a volatile day and a control subject trading on a calm day would differ for reasons unrelated to the treatment. Pre-programmed, identical price sequences remove both problems and make behavioral differences attributable to the treatment alone.

We recruit about 180 participants, 90 per condition, from finance students, CFA candidates, and junior investment professionals through Pitt's business school and local firms. Eligibility requires completion of at least one financial markets course, current CFA enrollment, or current employment in a finance role. Institutional portfolio managers would be a better sample but are not accessible at this scale, and the individual context they carry cannot be

controlled for. Ninety per condition provides sufficient power to detect a medium effect size (Cohen's d about 0.5) at 80% statistical power, grounded in the sample sizes of Guenther and Lordan (2023) at $n=193$ and Huang and Guenther (2024) at $n=132$. Total tournament payment cost is about \$450.

We pay on relative tournament performance, with the top 20% of earners receiving \$25 each, rather than on subjects' actual simulated trading profits, and the choice is deliberate. Paying trade earnings would reward following the correct AI recommendations and would tie payoff to the pre-programmed price path rather than to decision quality, confounding behavioral change with the quality of the advice, which is the same reason portfolio returns are not our primary outcome (Section IV). Tournament pay on relative performance mirrors the relative performance evaluation structures used in professional portfolio management and preserves the competitive, high-stakes environment that keeps the loss-aversion response emotionally live, without rewarding subjects for mechanically obeying a tool that is right 70% of the time. Flat participation fees, by contrast, would remove the motivation to maximize returns and with it the emotional engagement that makes the bias observable.

We use the direct response method rather than the strategy method because the disposition effect is an emotional response to price movements as they unfold. A strategy specified in advance bypasses that response entirely, turning a behavioral measurement into a stated preference measurement, and those are different things.

Random assignment handles selection bias. A cover story frames the study as a portfolio performance evaluation to reduce experimenter demand effects. A post-task manipulation check in the AI condition confirms subjects understood the recommendation came from an AI system rather than a human analyst; subjects who misunderstood are excluded, which is necessary

because the treatment effect requires AI attribution specifically. Dropout rates are monitored by condition throughout the study. All subjects are fully debriefed after data collection, informed of the true purpose of the study, and given the opportunity to withdraw their data. The cover story constitutes mild deception and will be reviewed and approved by the University of Pittsburgh IRB prior to data collection, consistent with standard ethical practice in behavioral economics research.

One design question is whether a banner notification produces genuine AI attribution or is dismissed as a routine pop-up. Logg et al. (2019) suggests receptivity to algorithmic advice is generally positive, but a more sophisticated interface with confidence scores or a brief rationale might produce stronger effects. Because added interface complexity introduces its own confounds, we will pilot the stimulus with a small group before the main study.

Two features of the current AI landscape bear directly on how far these results generalize. First, the configuration we study, a human who receives an AI recommendation and still decides whether to act, is not a hypothetical edge case but the dominant real-world configuration. AI decision support is already widespread across both institutional and retail investing, from algorithmic screening and robo-advisory products to AI research assistants that surface trade ideas for human review, so the human-in-the-loop setting the experiment isolates is the setting most investors already operate in. That makes the question timely and the result applicable to a large existing population rather than a speculative future one.

Second, and less comfortably, our estimate is conditional on a tool that is right about 70% of the time and on current levels of receptivity to algorithmic advice, and both are moving targets. As models improve in accuracy and as interfaces grow more persuasive, the size of the effect we measure could shift: plausibly upward if higher accuracy raises trust and compliance,

or downward if familiarity breeds the algorithm aversion documented by Dietvorst et al. (2015). We therefore read the design as estimating the effect of an in-the-moment recommendation at a realistic accuracy point for today's tools, not as a fixed constant. Treating tool accuracy and interface sophistication as parameters to be varied in follow-up work, rather than held at a single value, is the natural next step and the honest way to state what a single-accuracy result can and cannot tell us.

More broadly, institutional portfolio managers face pressures absent from our simulation: client mandates, regulatory constraints, tax rules, and career risk. Results therefore apply to finance-trained individuals in controlled investment tasks, and professional managers are a direction for future work rather than a design failure. The simulated environment also omits news flow, macroeconomic context, and liquidity constraints, which is consistent with prior work in this area (Weber and Camerer, 1998) and is a necessary condition for experimental control.

We use a between-subjects design because even with different price sequences in each arm, a within-subjects design would let subjects carry learned strategies across conditions: intuitions about when to trust AI recommendations, general heuristics about stock recovery patterns, and adapted risk tolerances. The treatment effect and the learning effect could not then be disentangled. Between-subjects eliminates the problem.

IV. Proposed Analysis

The primary outcome is PGR minus PLR calculated from all trading decisions across ten rounds. PGR is the number of gaining positions sold divided by the total number of opportunities to sell gaining positions, and PLR is the same for losing positions. A positive score indicates the disposition effect. The measure comes entirely from observed trading behavior, not self-report.

Portfolio returns are not the primary outcome because they would conflate behavioral change with AI advice quality. A subject who simply followed correct AI recommendations would show better returns regardless of whether their underlying decision-making pattern shifted. PGR minus PLR captures the pattern, not the outcome, which is the right thing to measure here.

The primary analysis is a one-sided two-sample t-test comparing mean PGR minus PLR between conditions, with a significance threshold of p less than 0.05. We also run an OLS regression of PGR minus PLR on the AI condition indicator with controls for financial literacy, years of finance training, and self-reported AI familiarity, which improves precision and allows for heterogeneous treatment effects.

In the AI condition, we calculate override rates separately for losing and winning position recommendations. Higher override rates specifically on losing positions would confirm that loss aversion is operating in the loss domain even when subjects have explicit AI guidance to sell, which is a direct behavioral test of the mechanism rather than of the aggregate score alone.

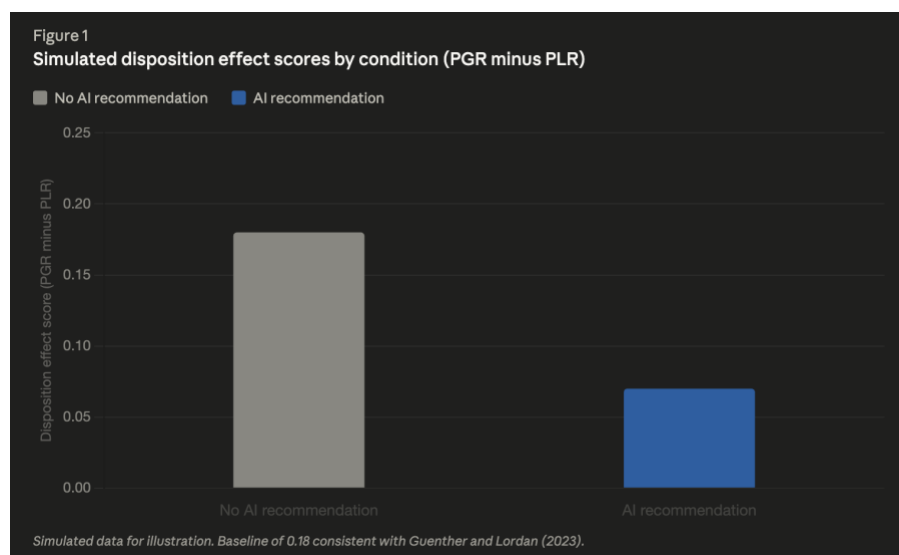


Figure 1 shows simulated data illustrating what a successful result looks like. The no-AI bar is about 0.18, consistent with Guenther and Lordan's (2023) baseline for professional-level

traders, and the AI bar is about 0.07. We expect partial rather than complete reduction: loss aversion does not vanish with AI guidance, but its magnitude should fall. The experiment succeeds if the AI condition shows a statistically significantly lower PGR minus PLR score at p less than 0.05, with override rates on losing positions lower in the AI condition than on equivalent positions in the no-AI condition.

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